

Lecture 3:

Supply and Demand

ECON101

Fall 2020



Supply Curve

$$Q_s = Q_s(P, DET^s)$$

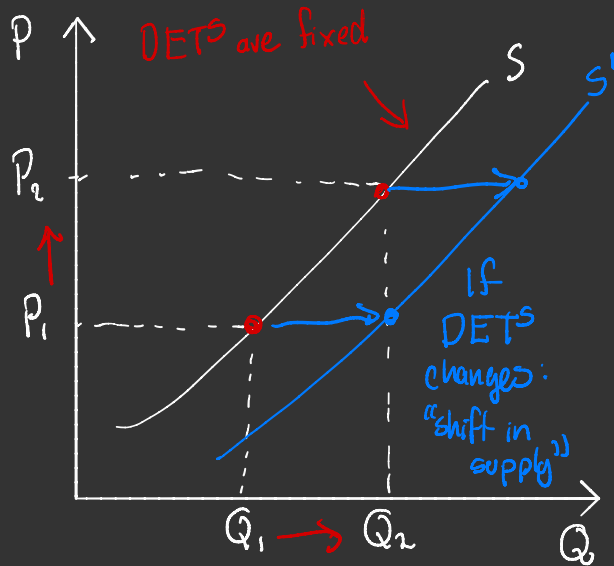
P = own price of the good
or service

DET^s = Determinants of the
supply

DET^s can be divided into
positive or negative determinants
depending on how they affect
the supply of the good

1. Production Costs : wages, rents, labor, etc, the higher costs are, the less quantity supplied.
2. Technology : technology used to produce the good, the more technology, the more the good can be supplied
3. Number of Sellers in the Market, the larger # sellers, the more the good can be supplied
4. Expectation of future prices : the future price of the good, or future prices of inputs

We can make a graph of $Q_s = Q_s(P, DET^s)$

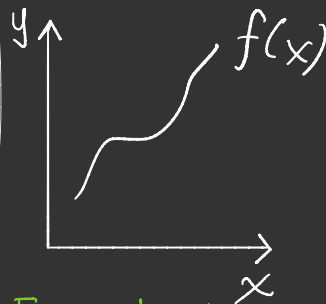


You can think about the supply as a schedule that represents the willingness to sell from the producers, i.e. the minimum price they are willing to accept to part with the good.

Supply is assumed to follow the "Law of the supply": the longer the prize, the greater the willingness of producers to sell.

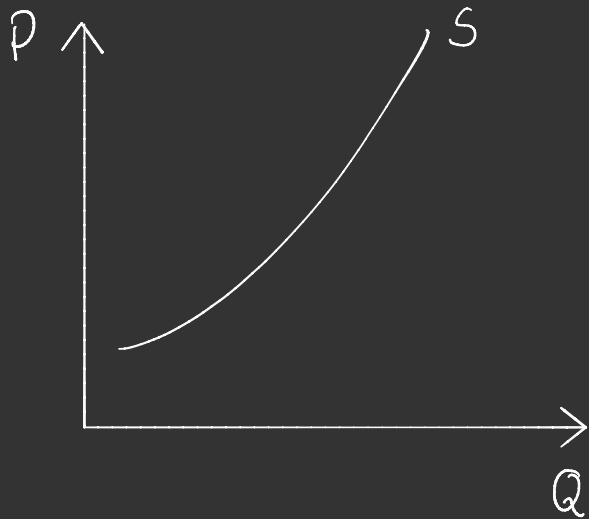
Movements along the supply curve due to changes in its own price are called changes in quantity supplied!

Note, in your math classes, for the function $y = f(x)$, you usually graph it:



For supply and demand diagrams, we invert the order!

- Changes due to price along the supply schedule are called **changes in quantity supplied**
 - Changes in the quantity supplied due to changes in the determinants DET^s, other than the price, are called **shifts in supply**
-



Supply schedule can have any shape, as long as it slopes upwards (Law of Supply).

A particular simple case to graph is the case of linear supply.

Linear Supply

$$Q_s = dP + DET^s, \quad d > 0$$

To graph it, solve for P :

$$\frac{Q_s - DET^s}{d} = P \quad \text{Eq. of the supply schedule (in the graph)}$$

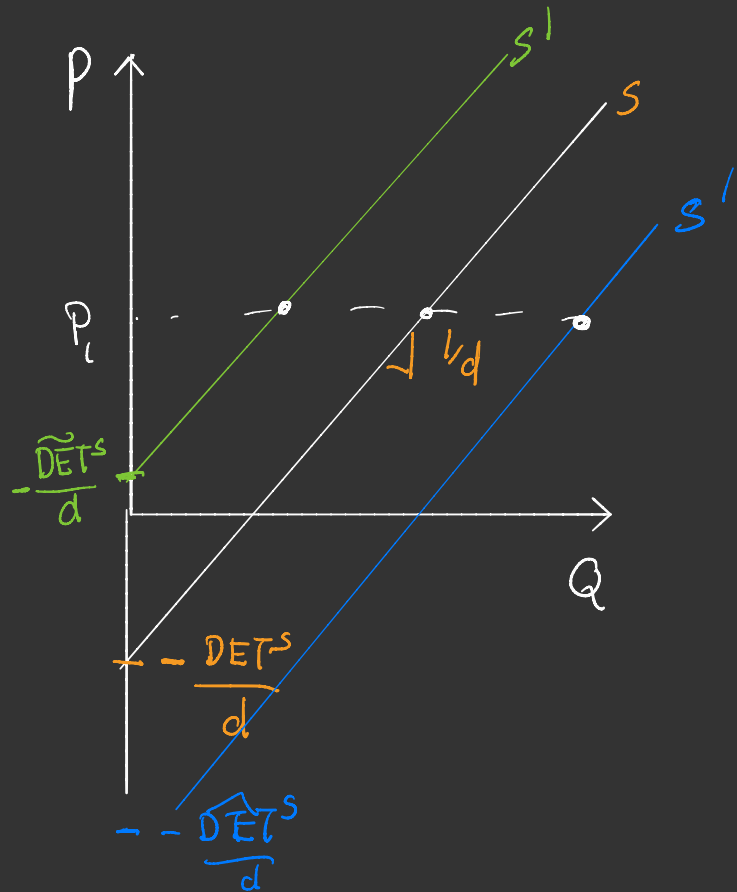
$$P = \frac{1}{d} \cdot Q_s - \frac{DET^s}{d}$$

slope: $\frac{1}{d}$

intercept: $-\frac{DET^s}{d}$

$DET^s \uparrow$

$DET^s \downarrow$



Demand Curve

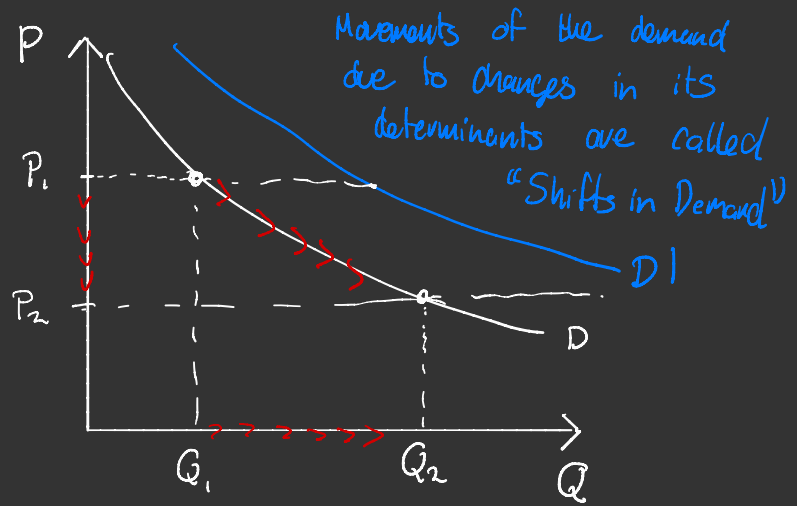
$$Q_D = Q_D(P, DET^D)$$

where P is the price of the good and DET^D are other variables that affect the quantity demanded.

You can think of the demand curve as schedule that represents how much consumers are willing to pay for some quantity of the good.

It follows the "Law of Demand":

The larger the price, the lower the quantity demanded.



Movements along the demand curve due to a change in price (own price) are called changes in quantity demanded.

DET^D : Positive or Negative determinants

1. Disposable Income
2. Price of Complementary or Substitute goods
3. Consumer Expectations of future price
4. Consumer Tastes

Complements and Substitutes

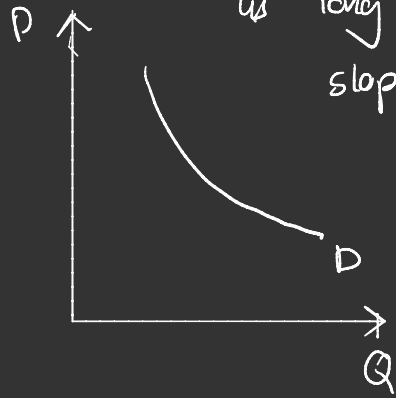
- Are determinants of the demand
 - $p^s \uparrow \rightarrow Q_D \uparrow$
 - $p^s \downarrow \rightarrow Q_D \downarrow$

} + Shifter

 - $p^c \uparrow \rightarrow Q_D \downarrow$
 - $p^c \downarrow \rightarrow Q_D \uparrow$

} - Negative Shifter

The demand schedule can have any shape
as long as it is downward
sloping.



A particularly simple example is the case of Linear Demand

Linear Demand Curve

$$Q_D = DET^D - bP, \text{ where } b > 0.$$

To graph it, solve for P :

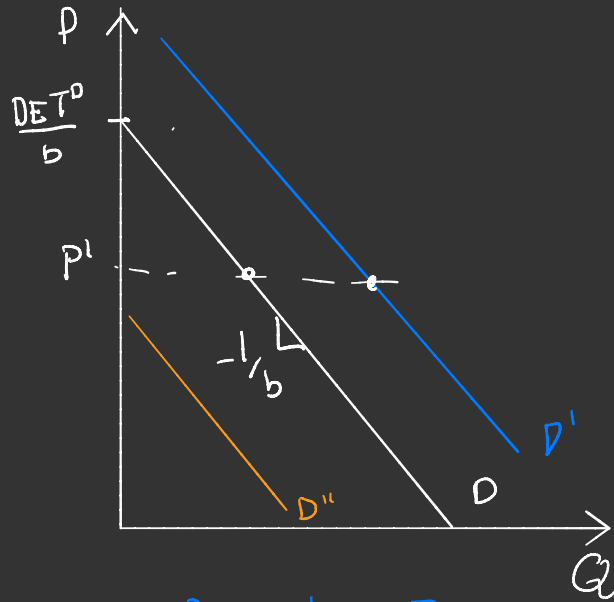
$$\frac{DET^D - Q^D}{b} = P$$

$$P = \frac{DET^D}{b} - \frac{1}{b} Q^D$$

Eq. of the graph of the demand

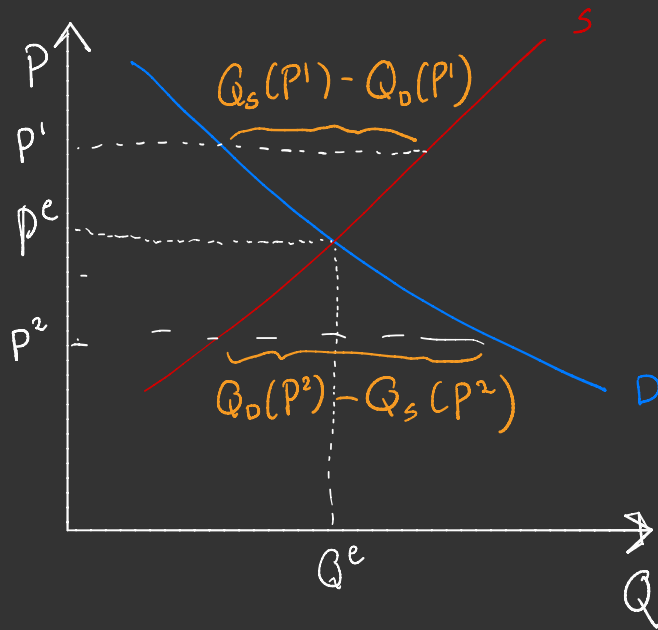
Intercept: $\frac{DET^D}{b}$

Slope: $-\frac{1}{b}$



$DET^D \uparrow$: Income Increased

$DET^D \downarrow$: $P^c \uparrow$



Equilibrium: $Q_D(P^e) = Q_S(P^e)$

while holding
the other
determinants
fixed.

$$Q^e = Q_D(P^e) = Q_S(P^e)$$

Market Equilibrium is the pair (P^e, Q^e)

Surplus (Excess Supply) : $P > P^e$

$$\text{Surplus} = Q_S(P) - Q_D(P)$$

Shortage (Excess Demand) : $P < P^e$

$$\text{Shortage} = Q_D(P) - Q_S(P)$$

For a linear demand & supply:

$$Q_D = DET^D - bP$$

$$Q_S = dP + DET^S$$

For equilibrium, we need that

$$DET^S + dP^e = Q^e = DET^D - bP^e$$

Solving for P^e yields

$$dP^e + bP^e = DET^D - DET^S$$

$$\Rightarrow P^e = \frac{DET^D - DET^S}{d + b}$$

To obtain Q^e , substitute P^e into either Q_D or Q_S :

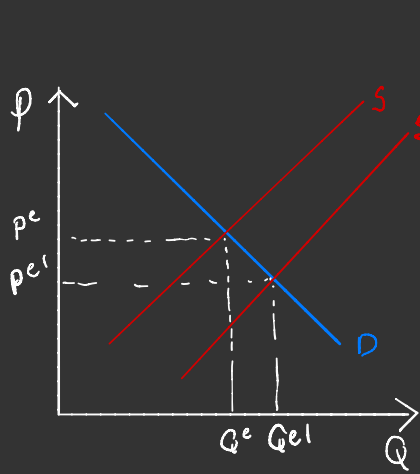
$$Q^e = DET^D - b \left(\frac{DET^D - DET^S}{d + b} \right)$$

$$Q^e = \frac{(d + b)DET^D - b(DET^D - DET^S)}{d + b}$$

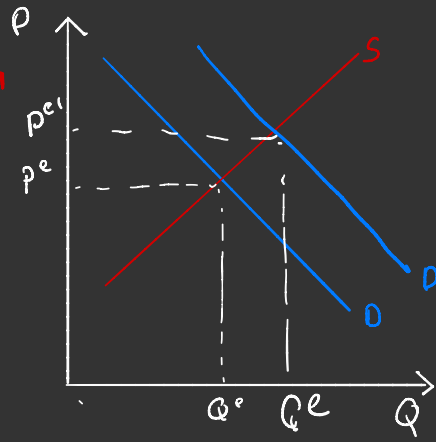
$$Q^e = \frac{d DET^D + b DET^S}{d + b}$$

What happens when the determinants change?

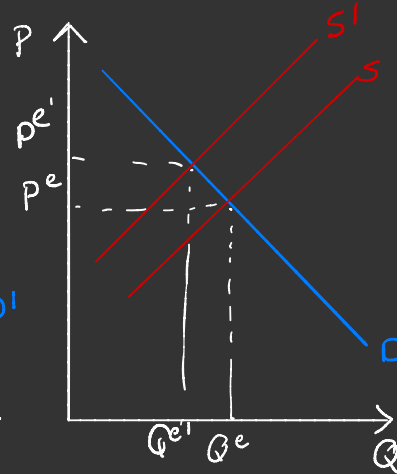
$$p^e = \frac{DET^D - DET^S}{b+d}; \quad Q^e = \frac{dDET^D + bDET^S}{b+d}$$



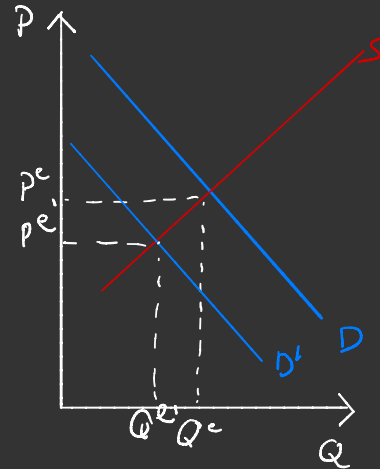
$DET^D \uparrow \Rightarrow S \uparrow$
 $p^e \downarrow, Q^e \uparrow$



$DET^D \uparrow \Rightarrow D \uparrow$
 $p^e \uparrow, Q^e \uparrow$



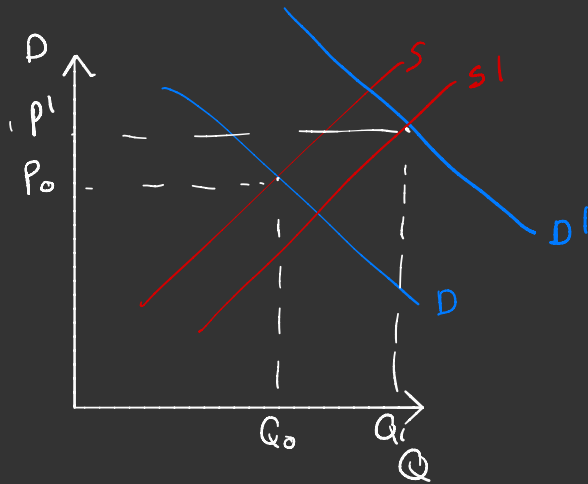
$DET^S \downarrow \Rightarrow S \downarrow$
 $p^e \uparrow, Q^e \downarrow$



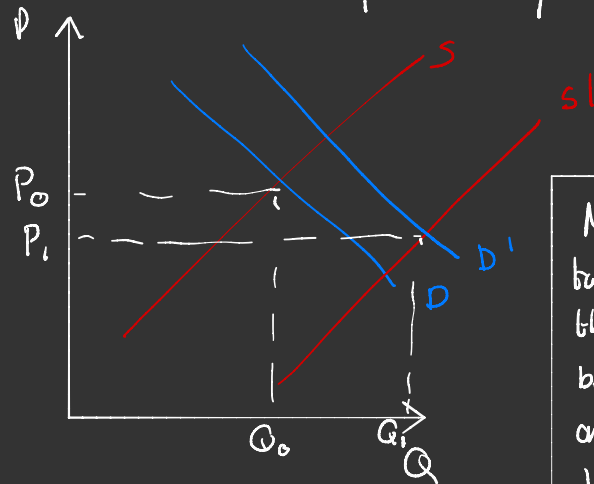
$DET^D \downarrow \Rightarrow D \downarrow$
 $p^e \downarrow, Q^e \downarrow$

$$p^e = \frac{DET^D \uparrow + DET^S \uparrow}{b+d} ; Q^e = \frac{d DET^D \uparrow + b DET^S \uparrow}{b+d}$$

$DET^S \uparrow$ $DET^D \uparrow \Rightarrow Q^e \uparrow ; p^e = ?$ We would need accurate info. about the magnitude of DET^D and DET^S to pin down precisely p^e .



$DET^D > DET^S$
 $Q^e \uparrow, p^e \uparrow$



$DET^D < DET^S$
 $Q^e \uparrow ; p^e \downarrow$

Moral: When the two curves shift at the same time, will be able to predict one of the eq. variables unequivocally, but the other one will be uncertain unless we have information on the magnitude of the change.

The market for webcams

Because of the pandemic, a lot of people had to work from home $\rightarrow$ you need a webcam.

On the other hand, most webcams are manufactured in China,

$\rightarrow$ When all factories stopped, production stopped

This models what happened in real life:

- Check webcam prices in Amazon and the price history
- Check availability!

